Geometry Honors — Midterm Examination

Topics: Logical Reasoning · Proofs · Parallel Lines · Triangles · Quadrilaterals · Transformations · Similarity · Right Triangles & Trigonometry · Circles · Area & Volume

Total Points: 150 **Time Allowed:** 180 minutes **Show all work.**

Name: ____________________ Date: _______________ Period: __________

Instructions: This exam has **5 sections** covering the Geometry Honors curriculum. Show all work for full credit. Justify answers using definitions, postulates, and theorems. A calculator is permitted for Sections IV–V. No electronic devices for Sections I–III.

Section	Problems	Points
I. Multiple Choice	1–15	30
II. Short Answer	16–25	30
III. Proofs & Constructions	26–30	30
IV. Applications & Word Problems	31–38	30
V. Challenge Problems	39–40	30
Total	**40**	**150**

Section I: Multiple Choice (30 points)

Circle the best answer. Each question is worth 2 points.

1. Which of the following is the **converse** of the statement "If two lines are parallel, then corresponding angles are congruent"?
   - A. If two lines are not parallel, then corresponding angles are not congruent.
   - B. If corresponding angles are congruent, then two lines are parallel.
   - C. If corresponding angles are not congruent, then two lines are not parallel.
   - D. If two lines are parallel, then alternate interior angles are congruent.
2. In $\triangle ABC$, $m\angle A = 52\,\text{deg}$ and $m\angle B = 68\,\text{deg}$. What is $m\angle C$?
   - A. $40\,\text{deg}$
   - B. $52\,\text{deg}$
   - C. $60\,\text{deg}$
   - D. $120\,\text{deg}$
3. Which set of side lengths does **not** form a triangle?
   - A. 3, 4, 5
   - B. 7, 10, 12
   - C. 1, 2, 3
   - D. 8, 8, 15
4. The contrapositive of "If a quadrilateral is a square, then it is a rhombus" is:
   - A. If a quadrilateral is a rhombus, then it is a square.
   - B. If a quadrilateral is not a square, then it is not a rhombus.
   - C. If a quadrilateral is not a rhombus, then it is not a square.
   - D. If a quadrilateral is a rhombus, then it is not a square.
5. Which transformation **does not** always preserve congruence?
   - A. Translation
   - B. Reflection
   - C. Rotation
   - D. Dilation with scale factor $k \neq 1$
6. In parallelogram $ABCD$, $m\angle A = (3x + 15)\,\text{deg}$ and $m\angle B = (2x + 5)\,\text{deg}$. Find x.
   - A. 16
   - B. 28
   - C. 35

D. 48

7. Two similar triangles have areas 36 cm^2 and 100 cm^2. If the shorter base of the smaller triangle is 6 cm, what is the corresponding base of the larger triangle?

   A. 10 cm

   B. 15 cm

   C. 18 cm

   D. 25 cm

8. In right $\triangle ABC$ with $\angle C = 90\,\text{deg}$, $AC = 9$, $BC = 12$. What is $\sin A$?

   A. $\frac{3}{5}$

   B. $\frac{4}{5}$

   C. $\frac{3}{4}$

   D. $\frac{5}{4}$

9. A central angle in a circle measures $120\,\text{deg}$. What is the measure of the **intercepted arc**?

   A. $30\,\text{deg}$

   B. $60\,\text{deg}$

   C. $120\,\text{deg}$

   D. $240\,\text{deg}$

10. The circumference of a circle is 16π. What is its area?

    A. 8π

    B. 16π

    C. 32π

    D. 64π

11. Which point of concurrency is the intersection of the three **perpendicular bisectors** of a triangle?

    A. Incenter

    B. Centroid

    C. Circumcenter

    D. Orthocenter

12. The volume of a sphere with radius 6 cm is:

    A. $36\pi\ \text{cm}^3$

B. 72π cm^3

C. 144π cm^3

D. 288π cm^3

13. Lines $l \parallel m$ and transversal t intersects them. If $\angle 1 = (5x - 10)\,\text{deg}$ and $\angle 2 = (3x + 20)\,\text{deg}$ are alternate exterior angles, find x.

    A. 10

    B. 15

    C. 17.5

    D. 20

14. What is the equation of a circle centered at $(-3, 4)$ with radius 7?

    A. $(x+3)^2 + (y-4)^2 = 49$

    B. $(x-3)^2 + (y+4)^2 = 49$

    C. $(x+3)^2 + (y-4)^2 = 7$

    D. $(x-3)^2 + (y+4)^2 = 7$

15. If $\triangle ABC \sim \triangle DEF$ and the ratio of corresponding sides is $3 : 5$, what is the ratio of their perimeters?

    A. $3 : 5$

    B. $9 : 25$

    C. $5 : 3$

    D. $25 : 9$

Section II: Short Answer (30 points)

Show your work. Each question is worth 3 points.

1. Simplify: $\cos^2\theta + \sin^2\theta =$ ______________

2. In the diagram, $\overline{AB}$ is tangent to circle O at point B, and radius $OB = 8$ cm. If $OA = 10$ cm, find the length of $\overline{AB}$.

3. A regular hexagon has side length 6. Find its apothem (exact form).

4. The coordinates of $\triangle ABC$ are $A(1,2)$, $B(4,2)$, $C(1,6)$. After reflecting over the y-axis, then translating 3 units right, what are the coordinates of A''?

5. In $\triangle PQR$, S and T are midpoints of $\overline{PQ}$ and $\overline{PR}$ respectively. If $QR = 14$, find ST.

6. Find the measure of an inscribed angle that intercepts an arc of 230 deg.

7. A right triangle has legs of length 5 and 12. An altitude is drawn to the hypotenuse. Find the length of this altitude (exact or decimal to 2 places).

8. The surface area of a right cylinder with radius 4 cm and height 10 cm is ____________________ cm^2.

9. Two secants are drawn from exterior point P to a circle, intercepting arcs of 110 deg and 50 deg. Find $m\angle P$.

10. A regular pyramid has a square base with side 8 and slant height 7. Find its lateral surface area.

Section III: Proofs & Constructions (30 points)

Each question is worth 6 points. Show all reasoning.

1. **Two-column proof.** Given: $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$.
   Prove: Quadrilateral $ABCD$ is a parallelogram.

2. **Paragraph proof.** Given: Point D is the midpoint of $\overline{BC}$ in $\triangle ABC$. Line AD is extended to E such that $\overline{AD} \cong \overline{DE}$. Segments BE and CE are drawn.
   Prove: $BE + CE > BC$.

3. **Indirect proof.** Prove: In a right triangle, each acute angle measures less than 90 deg.

4. **Construction.** Using only a compass and straightedge, describe the step-by-step construction of the perpendicular bisector of segment $\overline{AB}$. Then prove that your construction is correct.

5. **Coordinate geometry proof.** Given: Points $A(-2, 3)$, $B(4, 7)$, $C(8, 1)$, $D(2, -3)$.
   Prove that $ABCD$ is a parallelogram by showing both pairs of opposite sides are parallel. Then determine whether it is a rhombus, rectangle, or square, and justify your answer.

Section IV: Applications & Word Problems (30 points)

Show all work. Point values are indicated in brackets.

1. A ladder 15 ft long leans against a building. If the foot of the ladder is 9 ft from the base of the building, how high up the wall does the ladder reach? (4)

2. A tree casts a shadow 24 m long. At the same time, a 2 m pole casts a shadow 3 m long. How tall is the tree? (5)

3. A circular fountain has diameter 12 ft. A 3 ft wide walkway surrounds it. Find the area of the walkway in terms of π. (5)

4. A satellite dish is parabolic (modeled as a circular arc). If the dish is 8 ft across and 2 ft deep, find the radius of the circle that models the dish. (5)

5. The centroid of $\triangle ABC$ has coordinates $(4, 5)$. If $A = (1, 2)$ and $B = (7, 8)$, find the coordinates of C. (5)

6. A regular octagon is inscribed in a circle of radius 10. Find the area of the octagon to the nearest whole number. (5)

7. The volume of a cone is 120π in^3 and the height is 12 in. Find the slant height. (5)

8. In $\triangle ABC$, $\angle C = 90\,\text{deg}$. The altitude from C to $\overline{AB}$ divides $\overline{AB}$ into segments of length 3 and 12. Find the length of this altitude. (4)

Section V: Challenge Problems (30 points)

These problems require synthesis of multiple concepts. Show all reasoning. Each question is worth 15 points.

Problem 39. Geometry of a regular polygon and trigonometry.
A regular dodecagon (12-sided polygon) is inscribed in a circle of radius R.

(a) Derive a formula for the side length of the dodecagon in terms of R.

(b) Derive a formula for the area of the dodecagon in terms of R.

(c) If $R = 10$, find the area to the nearest tenth.

(d) Explain why the area of a regular n-gon inscribed in a circle of radius R approaches πR^2 as $n \to \infty$.

Problem 40. Comprehensive proof and application.
In $\triangle ABC$, $AB = AC$. Let D be a point on $\overline{BC}$. The circumcircle of $\triangle ABD$ intersects $\overline{AC}$ at E (distinct from A). The circumcircle of $\triangle ADC$ intersects $\overline{AB}$ at F (distinct from A).

(a) Prove that $BDEF$ is a cyclic quadrilateral.

(b) Prove that $AE = AF$.

(c) If $AB = AC = 10$ and $BC = 12$, find the radius of the circumcircle of $\triangle ABD$ when D is the midpoint of $\overline{BC}$.

(d) Show that $\overline{EF} \parallel \overline{BC}$.

— End of Examination —

Answer Key (For Teacher Use)

Section I: Multiple Choice (2 pts each)

1. **B** — Converse switches hypothesis and conclusion.
2. **C** — $180 - 52 - 68 = 60\,\text{deg}$.
3. **C** — $1 + 2 = 3$, triangle inequality fails ($a + b > c$ required).
4. **C** — Contrapositive negates and reverses: "If not a rhombus, then not a square."
5. **D** — Dilation with $k \neq 1$ changes size; not a rigid motion.
6. **B** — Consecutive angles supplementary: $3x+15+2x+5 = 180 \Rightarrow 5x = 160 \Rightarrow x = 28$.
7. **A** — Area ratio $36:100 = 9:25$; linear ratio $3:5$; base $= 6 \cdot \frac{5}{3} = 10$.
8. **B** — $AB = \sqrt{9^2 + 12^2} = 15$; $\sin A = \frac{12}{15} = \frac{4}{5}$.
9. **C** — Central angle measure equals intercepted arc measure.
10. **D** — $2\pi r = 16\pi \Rightarrow r = 8$; $A = \pi(8)^2 = 64\pi$.
11. **C** — Circumcenter is intersection of perpendicular bisectors.
12. **D** — $V = \frac{4}{3}\pi(6)^3 = 288\pi$.
13. **B** — $5x - 10 = 3x + 20 \Rightarrow 2x = 30 \Rightarrow x = 15$.
14. **A** — $(x - h)^2 + (y - k)^2 = r^2 \Rightarrow (x + 3)^2 + (y - 4)^2 = 49$.
15. **A** — Perimeter ratio equals linear (similarity) ratio $= 3 : 5$.

Section II: Short Answer (3 pts each)

1. 1 (Pythagorean identity: $\cos^2\theta + \sin^2\theta = 1$)
2. $AB = \sqrt{OA^2 - OB^2} = \sqrt{100 - 64} = \boxed{6\text{ cm}}$ (tangent $\perp$ radius)
3. Apothem $= 6 \cdot \cot(30\,\text{deg}) = \boxed{3\sqrt{3}}$
4. Reflect $A(1, 2)$ over y-axis $\to A'(-1, 2)$; translate 3 right $\to \boxed{A''(2, 2)}$
5. $ST = \frac{1}{2}QR = \boxed{7}$ (Midsegment Theorem)
6. $\frac{1}{2}(230\,\text{deg}) = \boxed{115\,\text{deg}}$ (Inscribed angle theorem)
7. $c = \sqrt{5^2 + 12^2} = 13$; altitude $= \frac{5\cdot12}{13} = \boxed{\dfrac{60}{13} \approx 4.62}$
8. $SA = 2\pi r(r + h) = 2\pi(4)(4 + 10) = \boxed{112\pi}\ \text{cm}^2$
9. $m\angle P = \frac{1}{2}(110\,\text{deg} - 50\,\text{deg}) = \boxed{30\,\text{deg}}$ (secant-secant angle)
10. Lateral area $= 4 \cdot \frac{1}{2}(8)(7) = \boxed{112}$

Section III: Proofs (6 pts each)

26. Draw diagonal $\overline{AC}$. Show $\triangle ABC \cong \triangle CDA$ by SSS. Then $\angle BCA \cong \angle DAC$, so $AD \parallel BC$ (alt. int. $\angle$s). Similarly $AB \parallel DC$. Two pairs of parallel sides $\Rightarrow$ parallelogram.

27. $\triangle ADB \cong \triangle EDC$ by SAS ($BD = DC$, $AD = DE$, vertical $\angle$s). So $AB = EC$. Similarly $AC = EB$. In $\triangle ABE$, $AB + AE > BE$. Substituting: $EC + AC > EB$. Since $AC > CD = \frac{1}{2}BC$, we get $BE + CE > BC$ by the triangle inequality.

28. Suppose an acute angle $\angle A \geq 90\,\text{deg}$. Then $\angle A + \angle C \geq 180\,\text{deg}$, so $\angle B \leq 0\,\text{deg}$, impossible. Thus $\angle A < 90\,\text{deg}$.

29. (1) Center compass at A, draw arc above and below $\overline{AB}$ with radius $> \frac{1}{2}AB$. (2) Same radius, center at B, draw arcs intersecting the first pair at points P and Q. (3) Draw line PQ. Proof: $AP = BP = AQ = BQ$ (same radius), so P and Q are equidistant from A and B, hence lie on the perpendicular bisector.

30. Slopes: $m_{AB} = \frac{7-3}{4-(-2)} = \frac{4}{6} = \frac{2}{3}$; $m_{CD} = \frac{-3-1}{2-8} = \frac{-4}{-6} = \frac{2}{3}$. So $AB \parallel CD$. Similarly $m_{BC} = m_{DA} = -\frac{2}{3}$. Parallelogram confirmed. $AB = \sqrt{36+16} = \sqrt{52}$, $BC = \sqrt{16+64} = \sqrt{80}$. Adjacent sides not congruent $\Rightarrow$ not rhombus/square. $m_{AB} \cdot m_{BC} = -\frac{4}{9} \neq -1 \Rightarrow$ not rectangle. Conclusion: **parallelogram only**.

Section IV: Applications

31. $\sqrt{15^2 - 9^2} = \sqrt{144} = \boxed{12 \text{ ft}}$

32. $\frac{h}{24} = \frac{2}{3} \Rightarrow h = \boxed{16 \text{ m}}$ (similar triangles)

33. $R = 9$, $r = 6$; Area $= \pi(81 - 36) = \boxed{45\pi \text{ ft}^2}$

34. $R^2 = 4^2 + (R-2)^2 \Rightarrow R^2 = 16 + R^2 - 4R + 4 \Rightarrow 4R = 20 \Rightarrow \boxed{R = 5 \text{ ft}}$

35. Centroid $= \frac{A+B+C}{3} \Rightarrow C = 3(4,5) - (1,2) - (7,8) = (12-8,\ 15-10) = \boxed{(4,\ 5)}$

36. Area $= \frac{1}{2} \cdot 8 \cdot 10^2 \cdot \sin(45\,\text{deg}) = 400 \cdot \frac{\sqrt{2}}{2} = \boxed{566}$

37. $\frac{1}{3}\pi r^2(12) = 120\pi \Rightarrow r^2 = 30$; slant $l = \sqrt{30+144} = \boxed{\sqrt{174} \approx 13.2 \text{ in}}$

38. Altitude $= \sqrt{3 \cdot 12} = \boxed{6}$ (Geometric Mean Altitude Theorem)

Section V: Challenge Problems

39. (a) Central angle $= 360/12 = 30\,\text{deg}$. By Law of Cosines: $s = \sqrt{R^2 + R^2 - 2R^2 \cos 30\,\text{deg}} = R\sqrt{2 - \sqrt{3}}$.

    (b) Area $= 12 \cdot \frac{1}{2}R^2 \sin 30\,\text{deg} = 12 \cdot \frac{1}{2}R^2 \cdot \frac{1}{2} = \boxed{3R^2}$.

    (c) With $R = 10$: Area $= \boxed{300}$.

    (d) Area of inscribed n-gon $= \frac{1}{2}nR^2 \sin(2\pi/n)$. Let $x = 2\pi/n$. As $n \to \infty$, $x \to 0$. Area $= \pi R^2 \cdot \frac{\sin x}{x} \to \pi R^2$ since $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

40. (a) $\angle ADB + \angle ADC = 180\,\text{deg}$ (linear pair). $\angle AEB = \angle ADB$ (same arc in first circle). $\angle AFC = \angle ADC$ (same arc in second circle). So $\angle AEB + \angle AFC = 180\,\text{deg}$. Since $\angle BEF = 180\,\text{deg} - \angle AEB$ and $\angle BFD = 180\,\text{deg} - \angle AFC$, we get $\angle BEF + \angle BDF = 180\,\text{deg}$, so $BDEF$ is cyclic.

(b) Since $BDEF$ is cyclic, $\angle FEB = \angle FDB$. Since $\triangle ABC$ is isosceles, $\angle ABC = \angle ACB$. From the circumcircles, $\angle FDB = \angle FAB$ and $\angle EDC = \angle EAC$. By symmetry of the isosceles setup, $\triangle AFE \cong \triangle AEF'$, giving $AE = AF$.

(c) D midpoint $\Rightarrow BD = 6$, $AD = \sqrt{10^2 - 6^2} = 8$. In $\triangle ABD$: $AB = 10$, $BD = 6$, $AD = 8$. Circumradius: $R = \frac{abc}{4K} = \frac{10 \cdot 6 \cdot 8}{4 \cdot 24} = \boxed{5}$.

(d) Since $AE = AF$, $\triangle AEF$ is isosceles with $\angle AEF = \angle AFE = \frac{180 - \angle A}{2} = \angle B$. So $\angle AEF = \angle ABC$, giving $\overline{EF} \parallel \overline{BC}$ (corresponding angles).